



solutions with integrated vision technology for industrial automation, including high-speed, high-resolution cameras for real-time inspection and analysis. The SICK Inspector P-Series is a flexible vision sensor with advanced image processing algorithms that provide real-time optical inspection.

MVTec HALCON is a powerful software library for machine vision applications. It integrates AI-based algorithms and deep learning models for enhanced performance in automated optical inspection. ISRA Vision offers advanced 3D and 2D inspection systems, combining AI and deep learning to improve defect detection and quality assurance in production lines.

Balvinder Singh, Director of JB Instruments, highlights, “the high precision AI smart camera JBM series with 60fps, industrial reliability, and SDK integration. The JBM-C camera offers flexible OCR, precise positioning, calibration, autofocus, and optional multi-focal lenses. The JBM-C line camera, with a 13kHz M72 mount and high-performance sensor, supports GigE data transmission and is compatible with GigE Vision and GenICam tools.”

Integration of artificial intelligence and deep learning. AI and deep learning are transforming AOI systems, enabling adaptive learning, improved defect detection, and enhanced accuracy. This technological advancement is crucial for manufacturers striving to uphold high-quality standards while optimising production efficiency. Prominent machine models exemplifying these capabilities include the Koh Young Zenith Series, which uses AI and deep learning for precise solder joint inspections; the Omron VT-S1080, designed for high-speed and accurate defect detection with minimised false positives; and the

Challenges in AOI implementation

Despite their advantages, AOI systems face challenges that require careful consideration by design engineers:

Cost vs benefit. Advanced AOI systems can be expensive, especially those incorporating AI and multi-spectral imaging.

Integration complexity. Seamless integration into existing production lines demands significant planning and customisation.

Data overload. AI-powered systems generate large volumes of data that require efficient management and analysis tools.

Viscom S3088 Ultra, optimised for high-volume PCB assembly inspections.

Other notable systems, such as the CyberOptics SE3000, Saki 3D AOI, Yamaha YSi-V, and MIRTEC MV-6 Series, leverage AI and deep learning to enhance real-time defect recognition, process control, and inspection precision. These advancements ensure improved inspection reliability and efficiency, particularly in high-speed, high-precision manufacturing environments.

Shift towards 3D AOI systems.

The adoption of 3D AOI systems is accelerating due to the increasing complexity of electronic components. These systems provide in-depth information essential for detecting defects in complex geometries. Key models include Omron VT-S1080, offering high-resolution imaging for detecting solder joint defects and board warpage; Koh Young Zenith, utilising patented triangulation technology for soldering and component placement inspections; and CyberOptics SQ3000, which integrates multi-sensor technology for comprehensive defect detection.

Other notable systems include Viscom S3088 Ultra, employing high-speed laser triangulation for PCB surface analysis; Saki 3Di, combining advanced 3D imaging with high-resolution cameras; Nikon X-Ray/3D AOI, merging X-ray and 3D technology for detecting hidden defects; Test Research TR7700Q,

enabling real-time defect detection; and Yamaha YSi-V, which uses laser triangulation for precise inspection. These systems leverage advanced 3D scanning, imaging, and analysis techniques to identify defects early in production, improving quality and reducing errors.

Rising demand for miniaturisation and precision. As electronic devices shrink in size, AOI systems play a pivotal role in inspecting micro-scale components with high accuracy. This trend is particularly prominent in industries like consumer electronics, medical devices, and aerospace, where there is little margin for error.

Camtek is a well-known player in the AOI market, specialising in developing inspection solutions tailored for the semiconductor and electronics industries. Their AOI systems effectively detect defects in wafers, PCBs, and various electronic components. Viscom is a leading provider of inspection systems within the electronics manufacturing sector. The company’s advanced AOI systems, equipped with cutting-edge imaging technology and software, ensure precise and accurate inspections of electronic components.

Expansion of AOI applications beyond electronics. AOI systems are increasingly adopted in automotive, medical devices, and aerospace industries to enhance quality control and meet rigorous standards. This expansion into new markets is